



Automotive AI Accelerator · Chiplet SoC

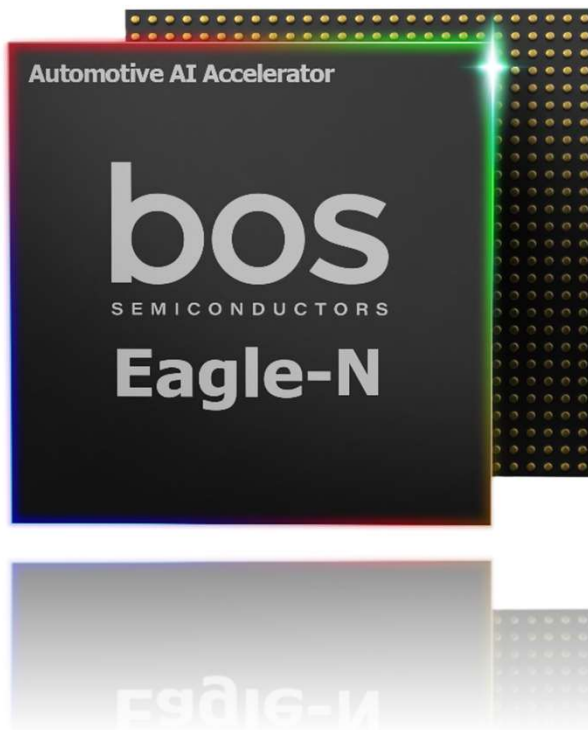
Eagle-N

A Modular, Chiplet-Based SoC
for Scalable Automotive AI

BOS Semiconductors · Hot Chips 2026

Eagle-N, High Level Overview

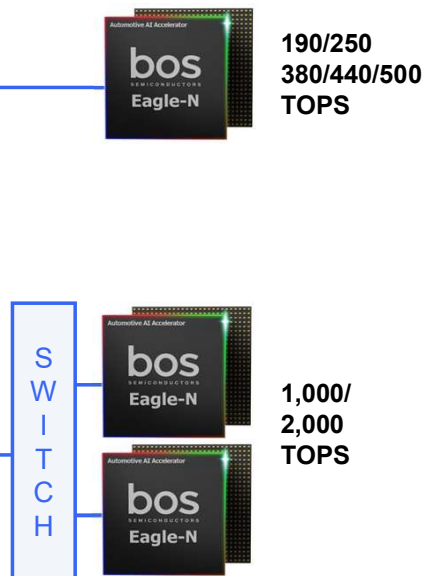
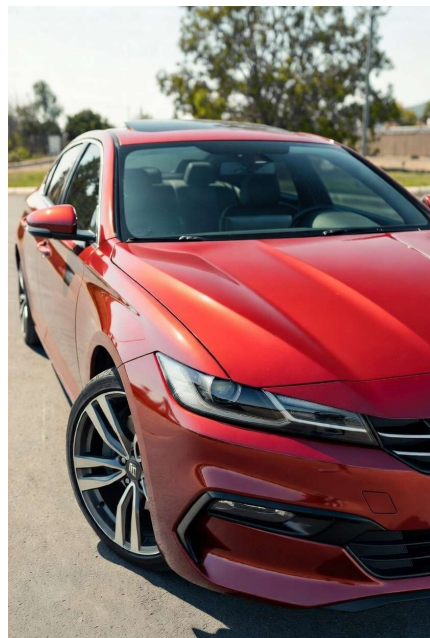
AI Accelerator for Automotive IVI/ADAS and Physical AI



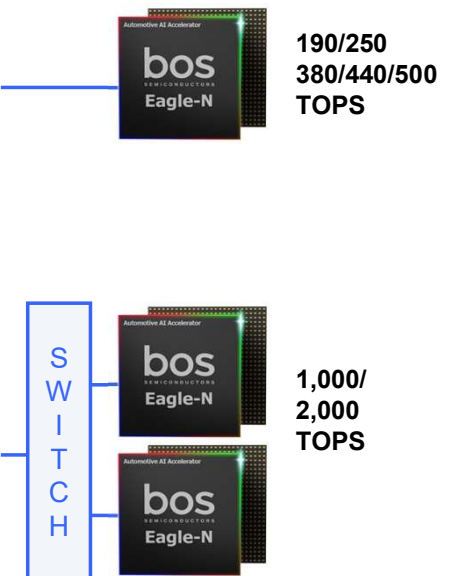
- 250 TOPS (Dense) of Tenstorrent NPU, Supports INT8, FP8, BF16, and Mx, Architecture is flexible to support future AI model
- Samsung Foundry 5nm Automotive process
- Multiple performance options through Harvesting and Chiplet
- Virtualization in NPU/SOC for multiple independent process
- Chiplet architecture for the flexible scalability and system expansion
- Communicate with main ADAS chip via PCIe/UCIe
- ARM Cortex A53 Quad core and LPDDR5X memory
- ASIL-B/AEC Q-100 Grade 2 (Safety Island, Security manager)

Scalability & Application

Flexible choice of NPU performance / Various application support where on-device AI is needed



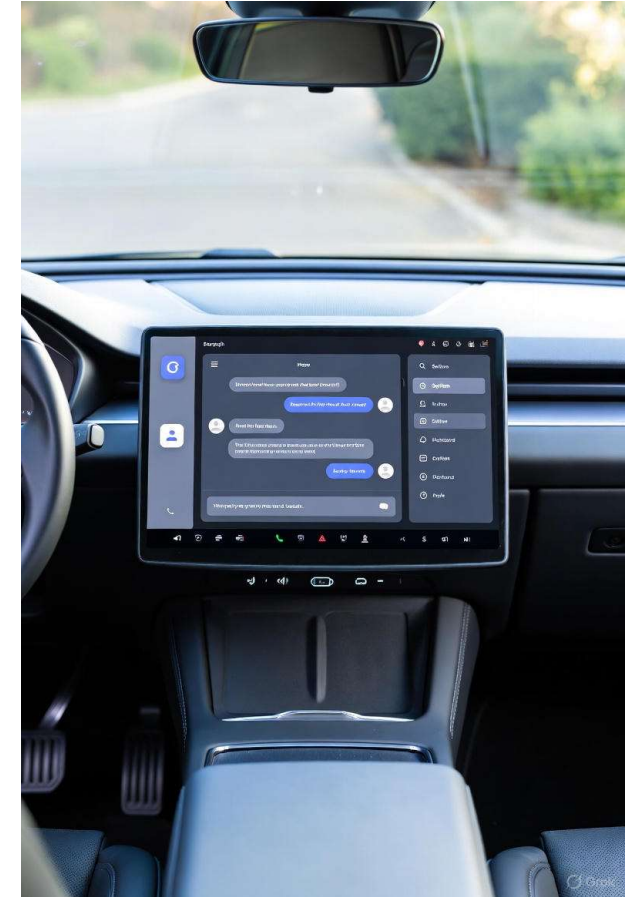
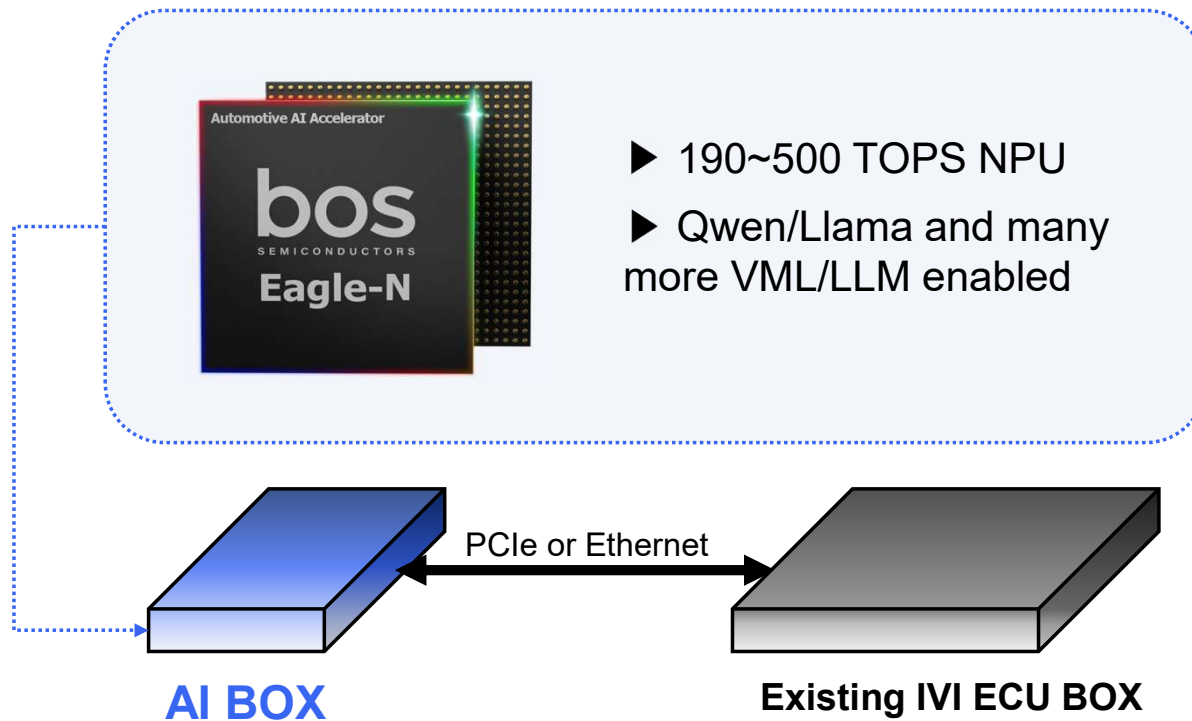
Automotive IVI and ADAS



On-device and Physical AI

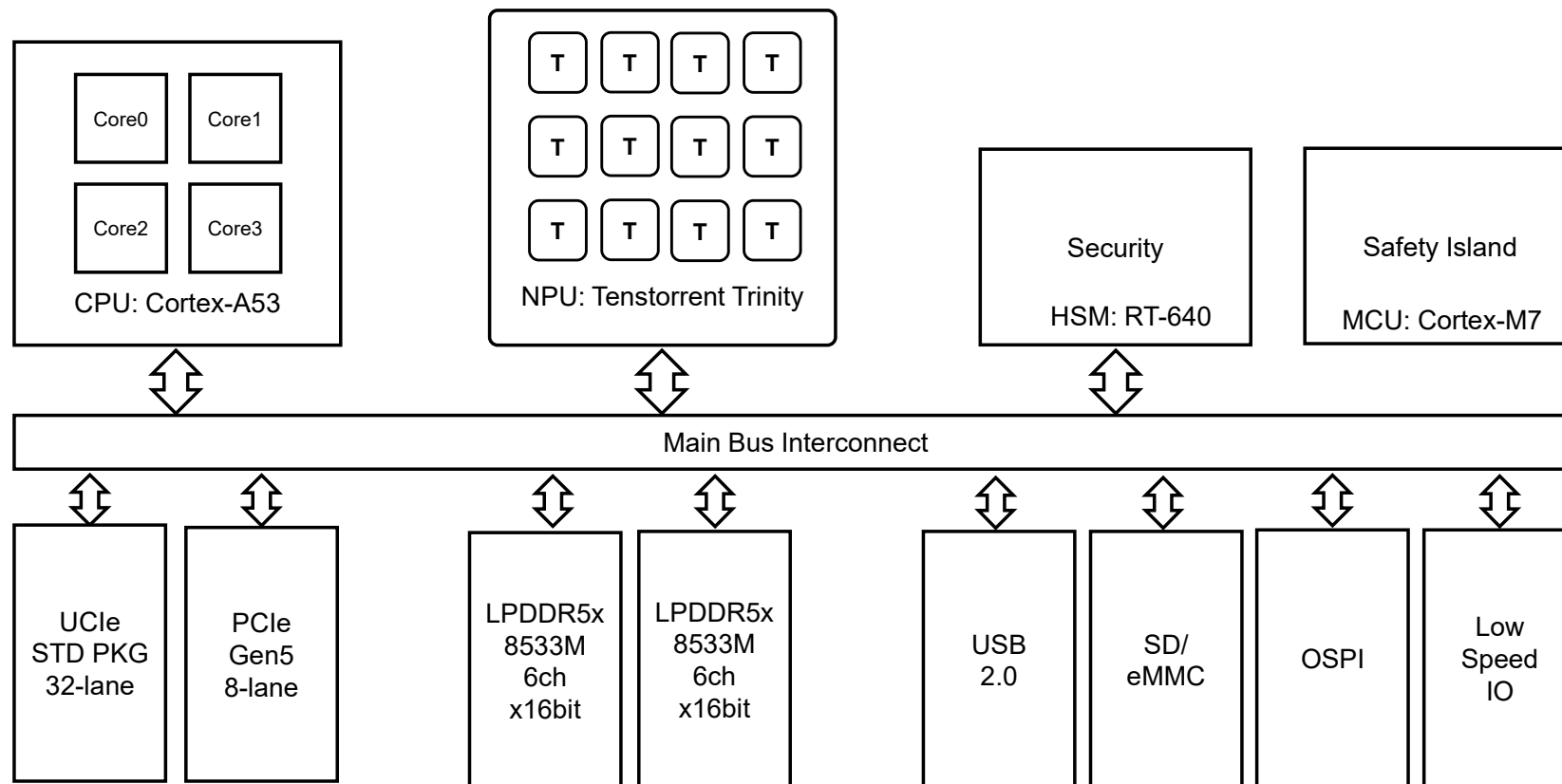
AI BOX ECU as AI Extension for IVI

Add 190–500 TOPS AI to existing IVI head units



Eagle-N Block Diagram

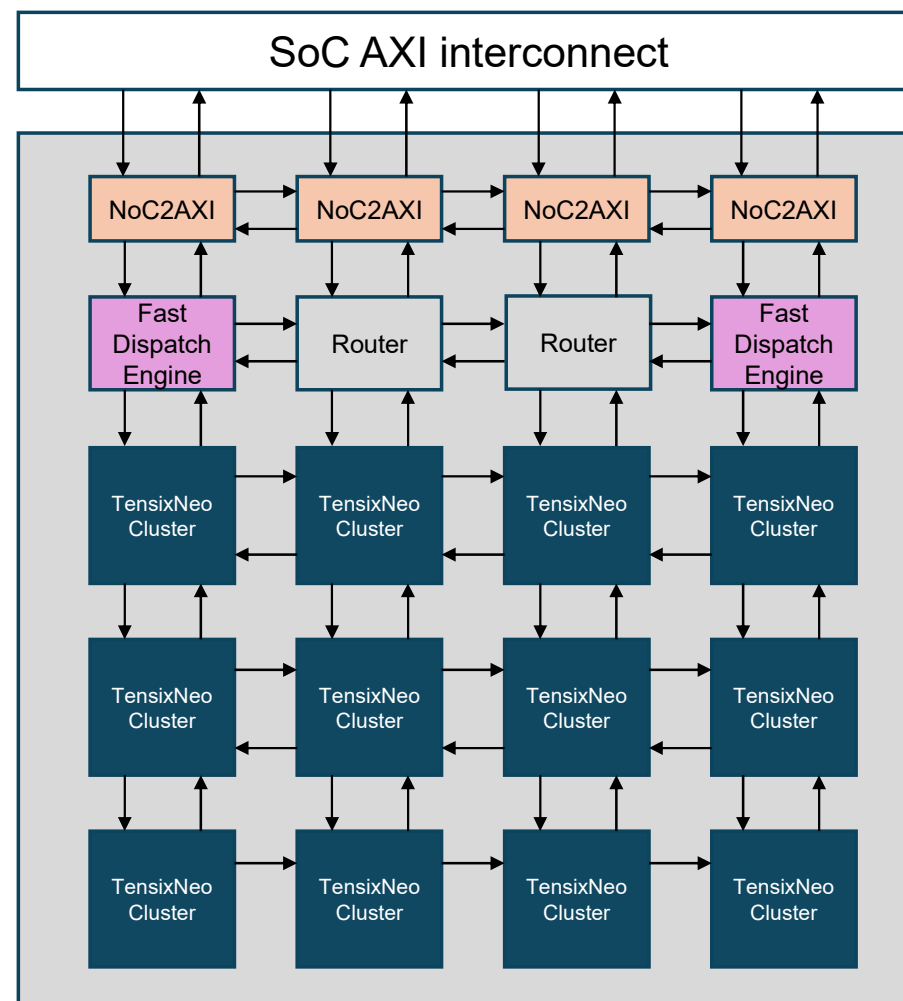
Cortex-A53 host · Tenstorrent Trinity NPU · security & safety islands



NPU Overview

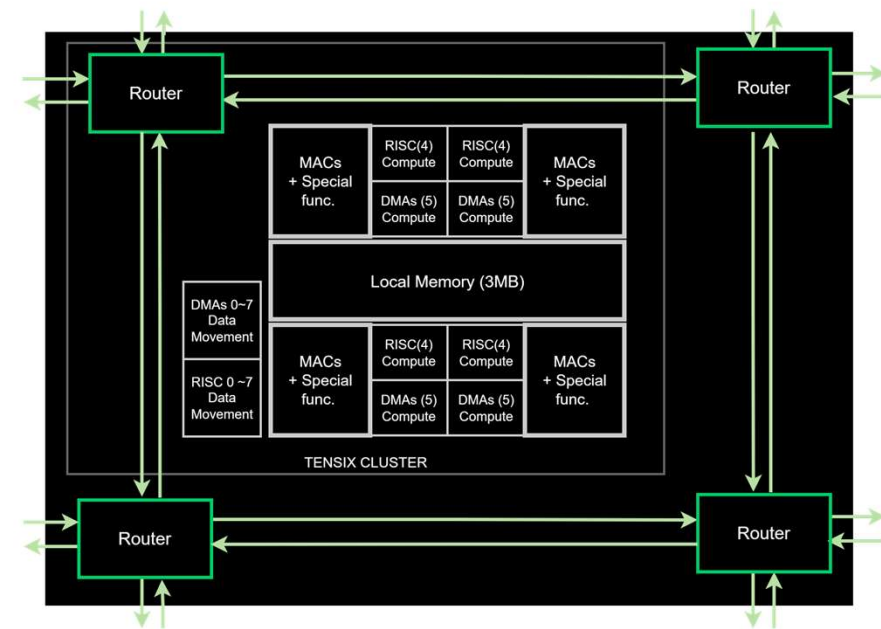
Tenstorrent Trinity IP — $12 \times$ Tensix Neo · 4×5 mesh NoC

- **Tenstorrent Trinity IP**
 - 12x Tensix Neo Cluster
 - 2x Fast Dispatch Engine
 - 2x Router
 - 4x NoC2AXI
- **NoC within NPU**
 - 256 Byte bidirectional NoC mesh
 - Both directions run simultaneously
 - NoC2AXI
 - 64 Byte Manager/Subordinate AXI ports
- **Fast Dispatch Engine**
 - Fast kernel dispatch to Tensix Neo clusters
 - Pre-fetch and queue kernels to mitigate Host-to-IP or DRAM-to-IP latency
 - Dispatch them to the clusters as soon as they are ready



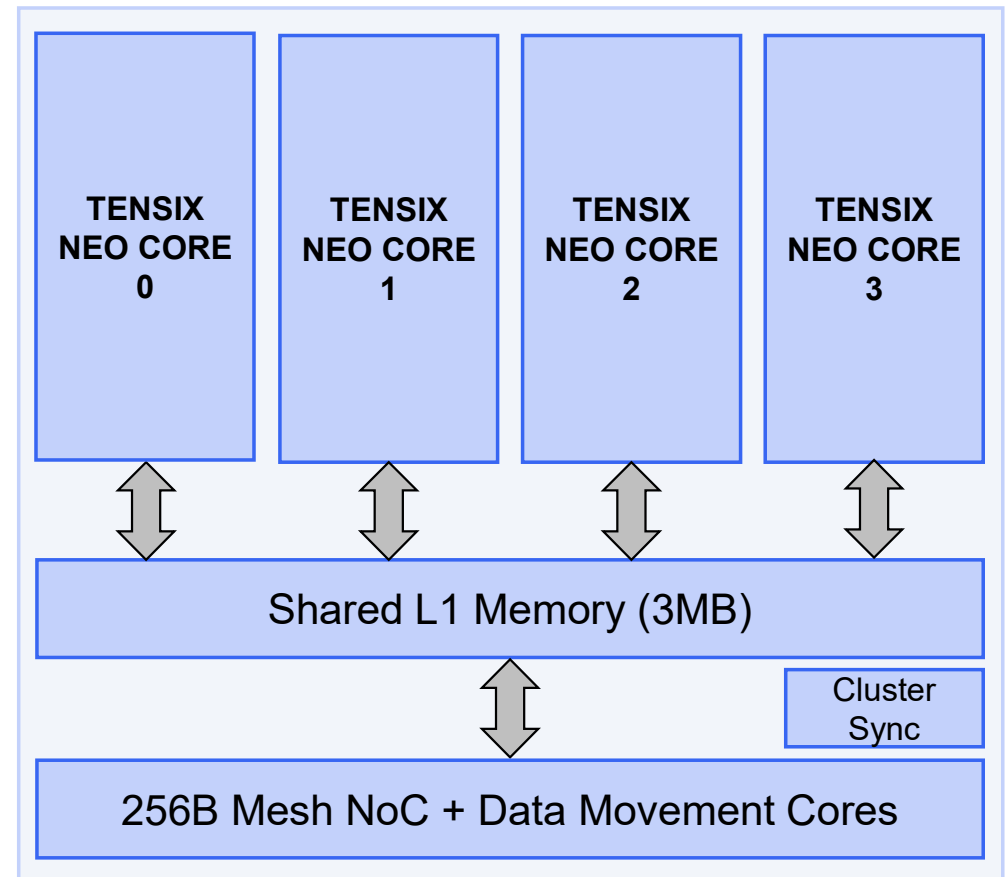
NOC Fabric for data movement

- **NOC**
 - **Bidirectional 256B Width Bus**
 - A tile can be receiving a weight data while it streams out a result without stalls
 - **4x5 2D Meshed NOC**
 - A 2D mesh gives every tile up to four neighbors (N/S/E/W) plus its local port
 - **Configurable Bidirectional Quad Port Router**
 - Flexible and diverse routing protocol
 - Adaptive NoC Routing for Diverse Workloads
 - Flexible On-Chip SRAM Utilization
 - **The NoC can be extended to route data over die-to-die interfaces**



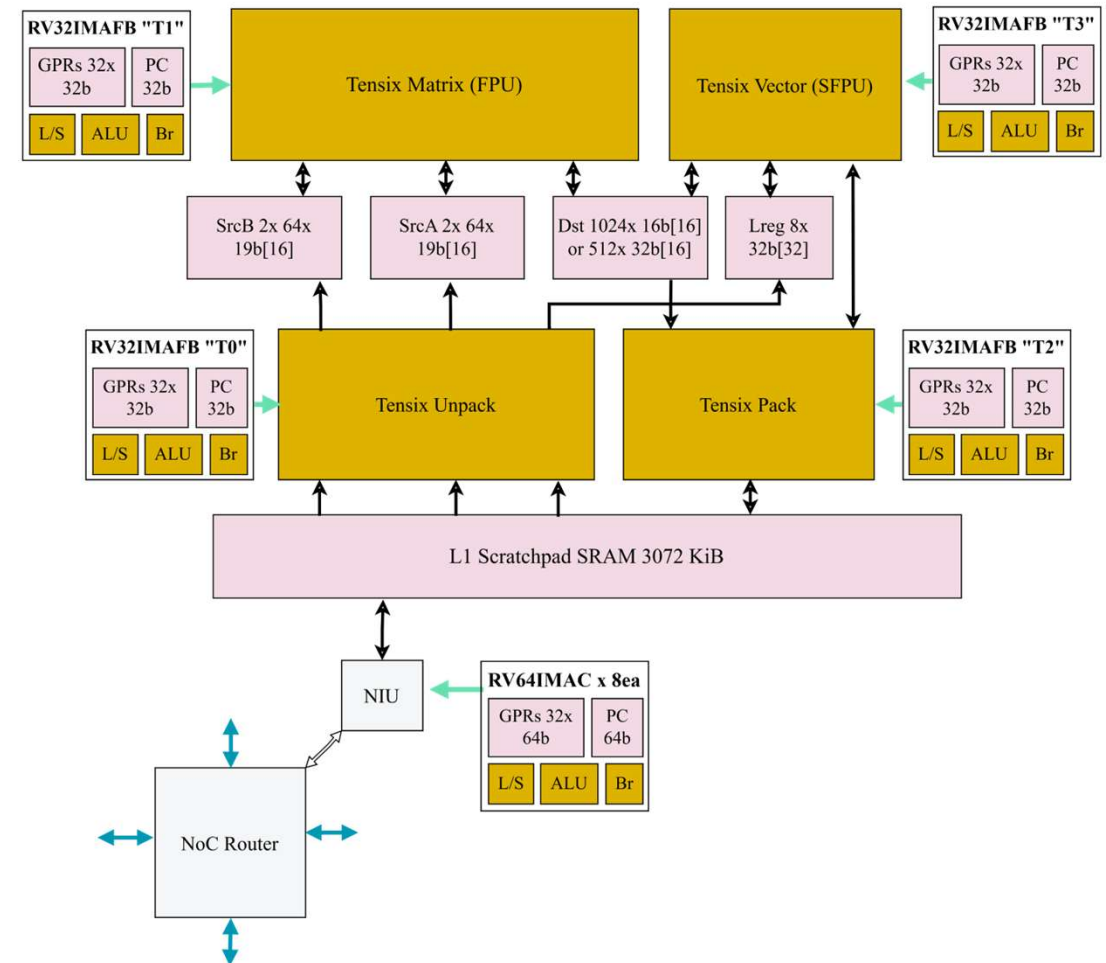
NPU Tensix Neo Cluster

- The compute block of the NPU
- Each cluster contains:
 - 4 compute engines
 - A NEO is the tensor-math core
 - A shared 3MB memory
 - Weights, activations, and intermediate tiles stay on-chip, cutting power-hungry DRAM round-trips for higher energy efficiency
 - A NoC Interface Unit (NIU)
 - Injection/ejection packet to/from the NoC
 - A data movement element (Overlay)
 - Prefetches into the register files
 - FPU's keep computing, sustaining near-peak MAC utilization



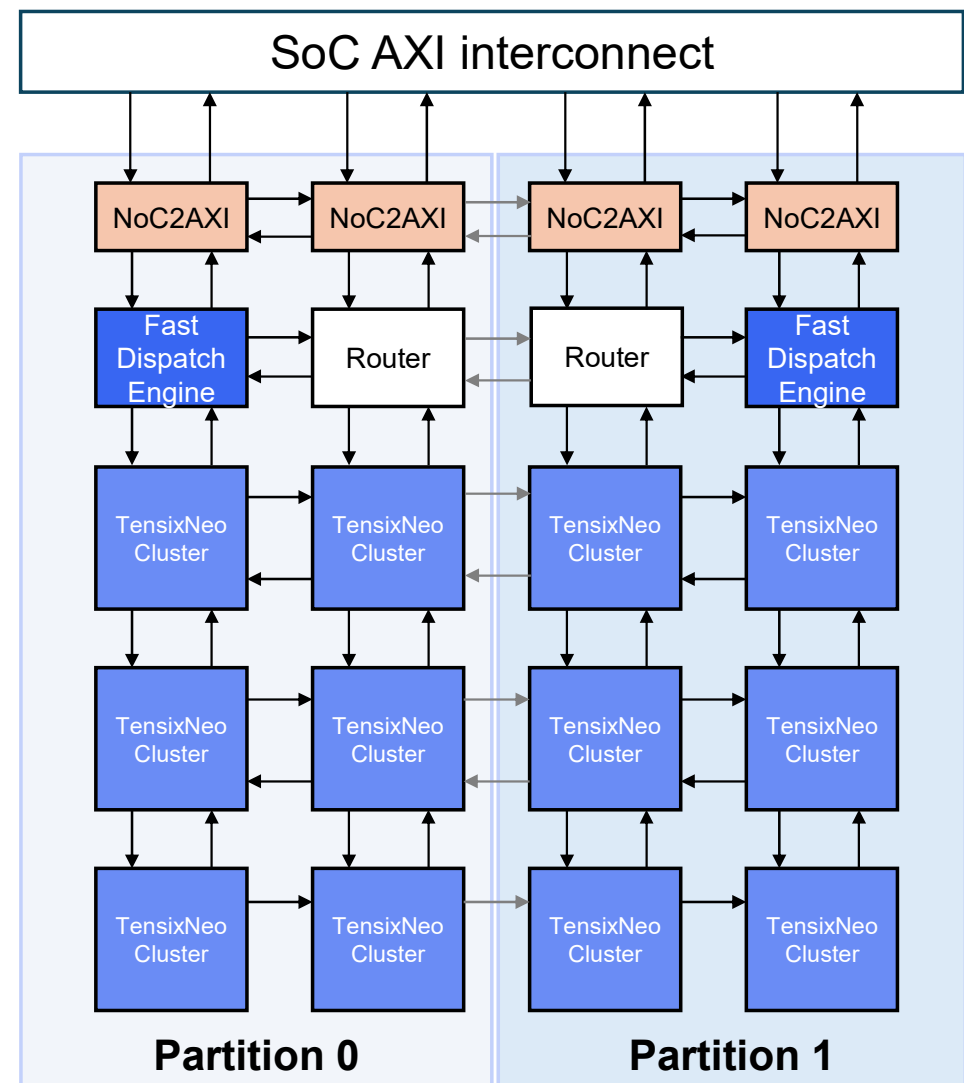
Tensix Core

- FPU for matrix multiplication, SFPU for general math operation such as exponential
- 4 RISC-V for controlling compute and 8 RISC-V for data movement
- Packer/Unpacker for data conversion between SRAM and register files
- 3M bytes SRAM in each tensix core enables near memory computing.
- Explicit data movement: data movement is explicit and decoupled from compute.



NPU Virtualization

- **Virtualization**
 - Run one physical Trinity die as two independent logical NPUs
- **Dual Fast Dispatch**
 - Independent issues to its own 6 Tensix clusters via the dedicated hardware interface
- **Independent partition**
 - Each partition has its own compute, dispatch, DRAM/AXI I/O, clocks, and reset — able to run a different workload independently
- **Memory access control**
 - SoC interconnect supports MMU for memory protection between the two partitions



OCA (Open Chiplet Atlas) for Chiplet Systems

- **Benefit of Chiplet Systems**

- Yield improvement, Reticle restriction, Heterogeneous process, Reusability

- **Issues in Chiplet Systems**

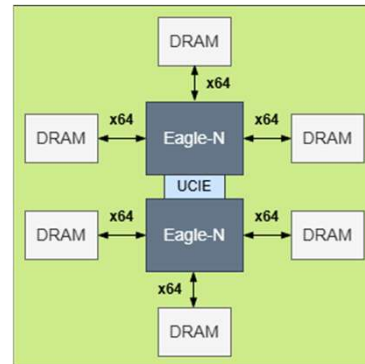
- Chiplet discovery and enumeration
- System boot and initialization
- Inter-chiplet messaging, interrupt delivery, Time synchronization
- Memory addressing and access control
- RAS (Reliability, Availability, Serviceability)
- Functional Safety and Security management
- DFT and Test architecture

- **OCA defines a common system architecture for Chiplet systems**

Eagle-N OCA Support

- **Chiplet configuration using Eagle-N**

- Eagle-N + Eagle-N:

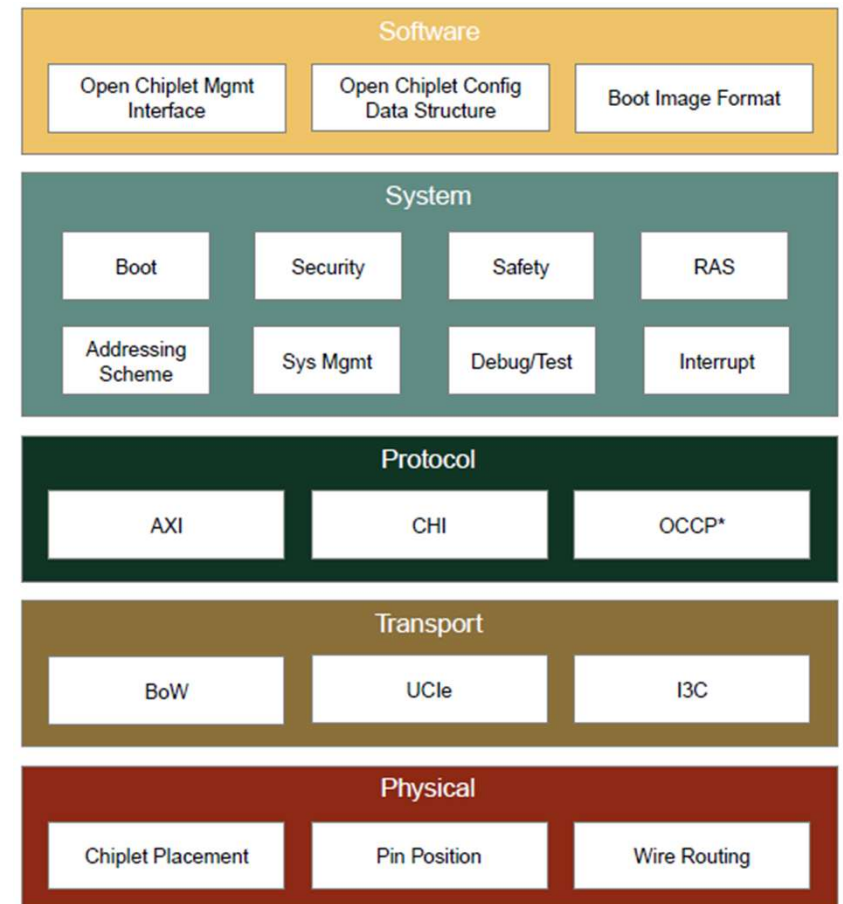


- Host SoC + Eagle-N:



- **Eagle-N is designed for compatibility with OCA.**

- Transport layer: UCle, I3C
 - Protocol layer: AXI, OCCP
 - System layer

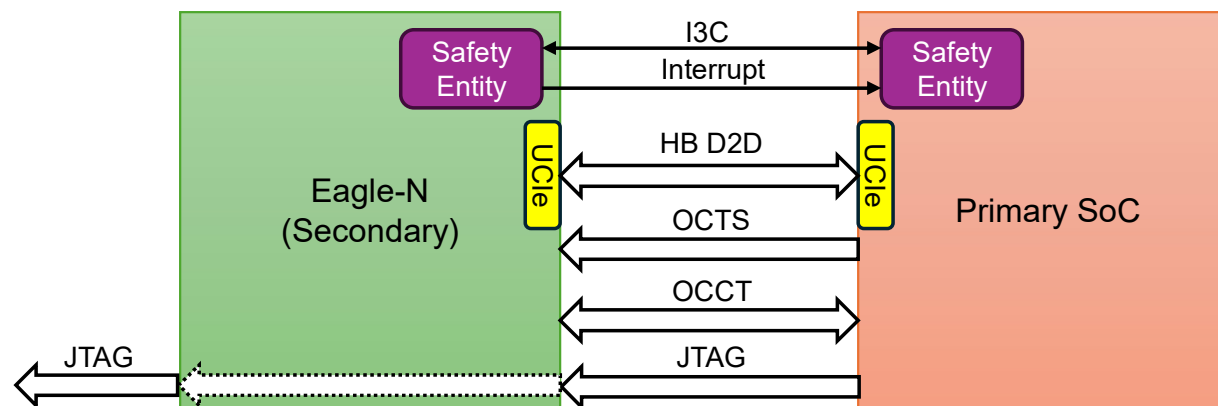


* OCCP: Open Chiplet Configuration Protocol

Layered view of OCA Chiplet Interoperability

Protocol and Transport Layer

Protocol	Transport	Remark
High Bandwidth D2D	UCle Standard Package	OCA AXI over UCle
Low Bandwidth D2D	I3C	
Interrupt	Message Signaled Interrupt (MSI)	Doorbell in UCle block
Open Chiplet Time Synchronization (OCTS)	TMR_SYNC_LOAD, TMR_CNT_CREDIT	System timer sync signals
Open Chiplet Cross Triggering (OCCT)	CT_Req_out, CT_Ack_in, CT_Req_in, CT_Ack_out	One cross trigger port
JTAG	Additional JTAG bypass port	STAP (Secondary TAP)
Functional Safety	A dedicated I3C for functional safety message channel Two interrupts for functional safety	Fatal / Non-Fatal



*OCTS: OpenChiplet Time Synchronization
*OCCT: OpenChiplet Cross Triggering

System Layer

Functional Domain	Description
Addressing	The UCle AXI Bridge integrates an Address Protection Unit (APU) on both the inbound and outbound paths, providing address translation and access protection for all transactions.
Boot	Eagle-N supports operation as both a primary chiplet and a secondary chiplet . When operating as a secondary chiplet, its firmware can be downloaded from the primary chiplet over the D2D interface.
System Time Management	An integrated 64-bit system time counter supports Open Chiplet Time Synchronization (OCTS). The counter can be configured to operate as either the primary counter or the secondary counter.
Security	Eagle-N provides secure interoperability with OCA-compliant chiplets through authenticated boot, lifecycle-based security, and AXI-based access control.
Interrupts	The UCle block supports MSI-based inter-chiplet interrupts through a built-in doorbell mechanism. The Safety Entity supports dedicated Fatal and Non-Fatal Error/Status interrupt signaling between chiplets.
Design for Debug	Eagle-N implements one OCCT channel for cross-trigger signaling and provides full OCCT connectivity as a leaf chiplet. ETR trace data can be transferred across the UCle channel and dumped into remote memory.
Design for Test	IEEE 1838 STAP is implemented to support daisy-chained ATE-mode BIST execution and scan dump across multiple dies. Open Chiplet Packetized Scan (OCPS) is not implemented.
Functional Safety	Eagle-N supports a one-to-one Safety Primary/Secondary Chiplet configuration and OCA-based FuSa messaging , including periodic channel checks and CRC.

Functional Safety overview

- **Random Hardware Fault Capability**
 - NPU domain: ASIL B
 - $\text{SPFM} \geq 90\%$, $\text{LFM} \geq 60\%$, $\text{PMHF} \leq 60 \text{ FIT}$
 - Safety Island: ASIL D
 - $\text{SPFM} \geq 99\%$, $\text{LFM} \geq 90\%$, $\text{PMHF} \leq 3 \text{ FIT}$
- **Systematic Fault Capability**
 - NPU domain & Safety Island: ASIL D
- **Deterministic Fault Handling**
 - Fault detection synchronized with the NPU inference cycle
 - 33 ms FHTI (aligned with the video frame period, 30fps), and 1 ms safety reaction time
- **Safe State**
 - Explicit failure report to system; or initiates a chip reset.

* **SPFM** : Single Point Fault Metric

* **LFM** : Latent Fault Metric

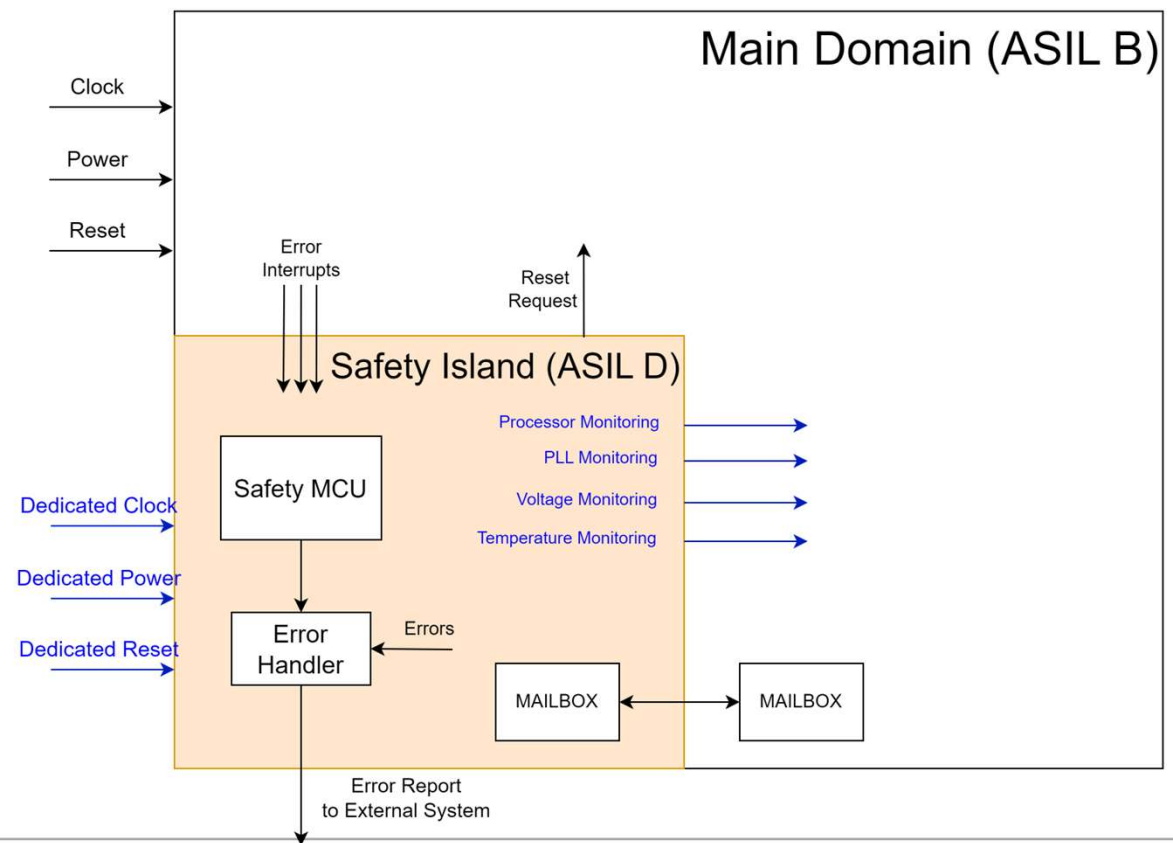
* **PMHF** : Probabilistic Metric for random Hardware Failures

* **FHTI** : Fault Handling Time Interval

Safety Island for Functional Safety

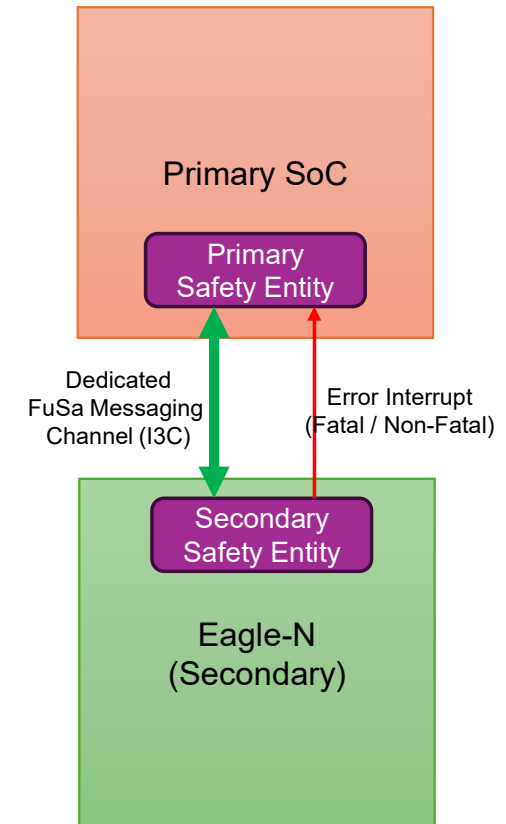
An independently isolated Safety Island keeps monitoring and reporting safety issues even when the Main Domain fails

- Power, Reset, Clock, NoC Bus, Memory, and Peripherals are fully isolated, enabling robust safety monitoring even in the event of Main Domain failure.
- Performs processor, temperature, clock, and power monitoring that cannot be reliably handled by the Main Domain itself.
- Its independence from the Main Domain enables fail-operational capability even when the Main Domain fails.



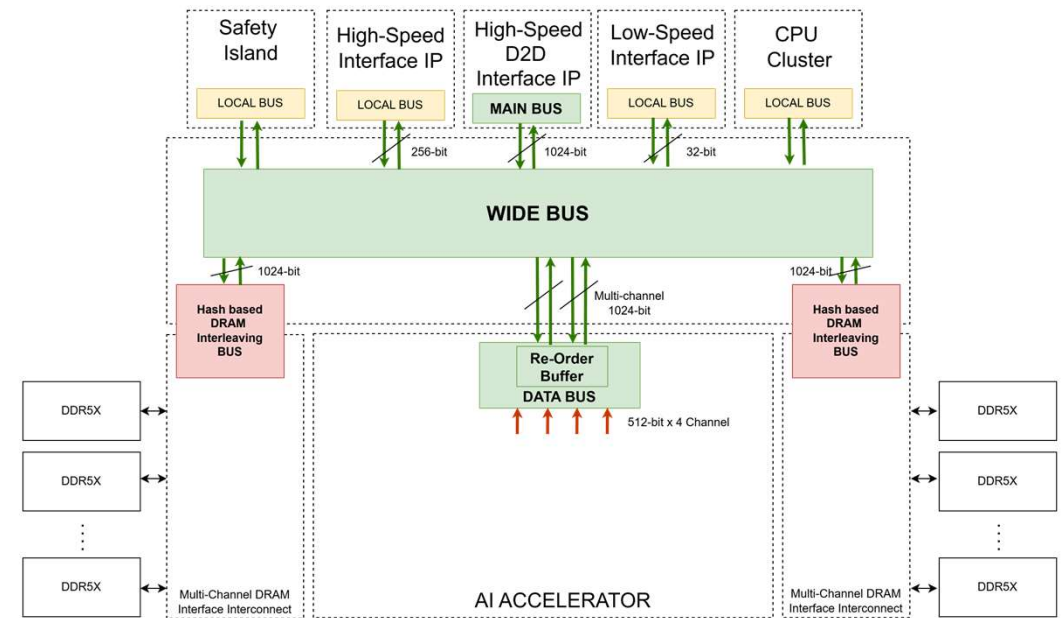
Functional Safety for OCA

- Chiplet Safety Protocol (CSP) with E2E protection
- Safety Entity in each Chiplet
 - Safety Island that keeps monitoring and reporting safety issues
- Hierarchical Safety Entities (Primary and Secondary)
 - Primary aggregates error info and determines the SiP-wide safe state
- Dedicated Error Interrupts : Fatal / Non-Fatal
- Dedicated FuSa Messaging Channel (I3C)
 - Executes error query, response, clearing, and self-test
 - Primary executes periodic self-tests through the Channel



Eagle-N Interconnect Architecture

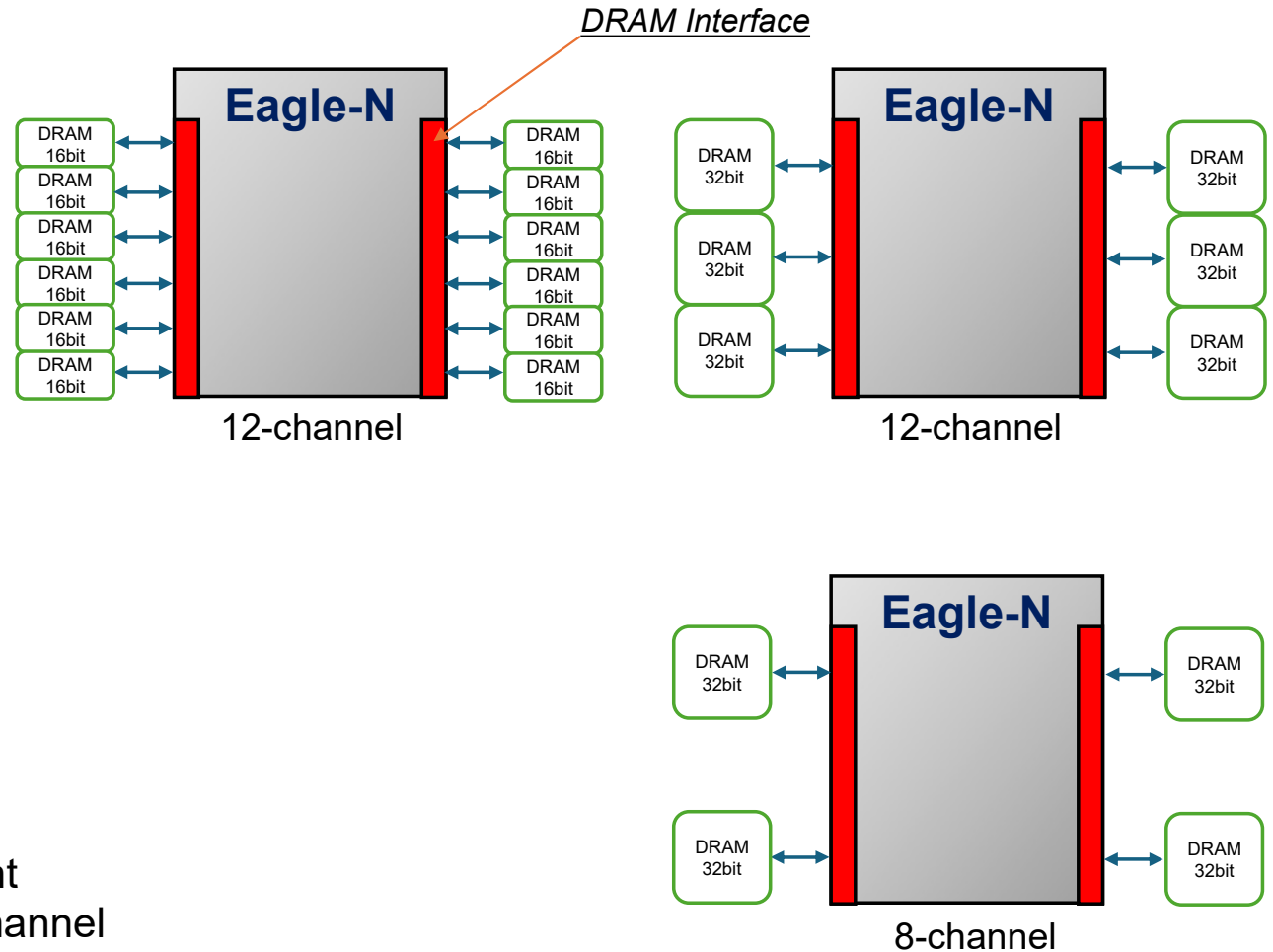
- **High bandwidth requirements**
 - Memory bandwidth decides AI model performance rather than compute performance in many AI models
- **Architecture Overview**
 - Hierarchical NoC architecture consisting of a high-bandwidth main fabric and subsystem-specific local buses, enabling scalable heterogeneous subsystems.
- **High-Bandwidth Interconnect**
 - 1024-bit @ 800 MHz (dual-channel) main fabric and 512-bit @ 1 GHz (four-channel) on-chip NPU fabric.
- **Memory Subsystem**
 - Up to 204 GB/s memory bandwidth through 12-channel LPDDR5X with channel interleaving.
- **Efficiency Optimization**
 - High bandwidth utilization under burst workloads through reorder buffers and transaction scheduling.



Interconnect Architecture

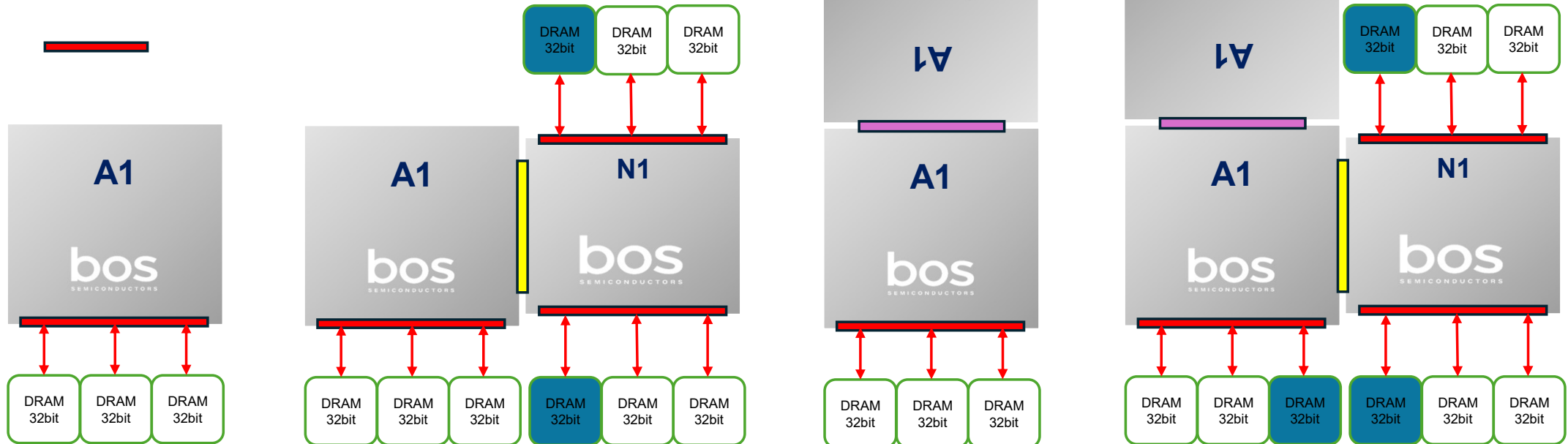
DRAM Interface

- **LPDDR5/5x up to 8533Mbps**
 - 16bit 12-channel
 - Bandwidth: up to 204 GB/s
- **Depopulation support**
 - 12-channel
 - 8-channel
 - 4-channel
- **DRAM Capacity**
 - Max. 8 GB per 1-channel
 - Up to 96 GB
- **DRAM interleaving**
 - 4KB interleaving for left and right
 - 256byte interleaving for each channel



DRAM Placement Example

- DRAM depopulation support
 - Eagle-A: 6ch → 4ch or 2ch
 - Eagle-N: 12ch → 8ch or 4ch

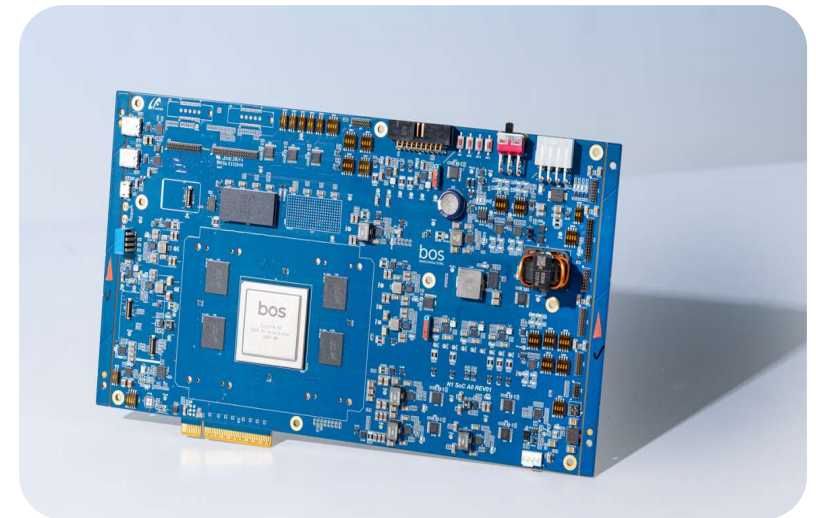


Eagle-N A0 sample for PoC

	Eagle-N A0	Eagle-N	Remark
NPU Performance	64 TOPS	250 TOPS	Effective fivefold performance improvement
DRAM Bandwidth	120 GB/s	204 GB/s	
DRAM Capacity	64 GB	96 GB	
PCIe	2x Gen5 4-lane 32 GT/s	1x Gen5 8-lane 32 GT/s (4+4 bifurcation)	
UCIe	None	Standard Package 2-module 16GT/s (total 32 pin)	64 GB/s per direction

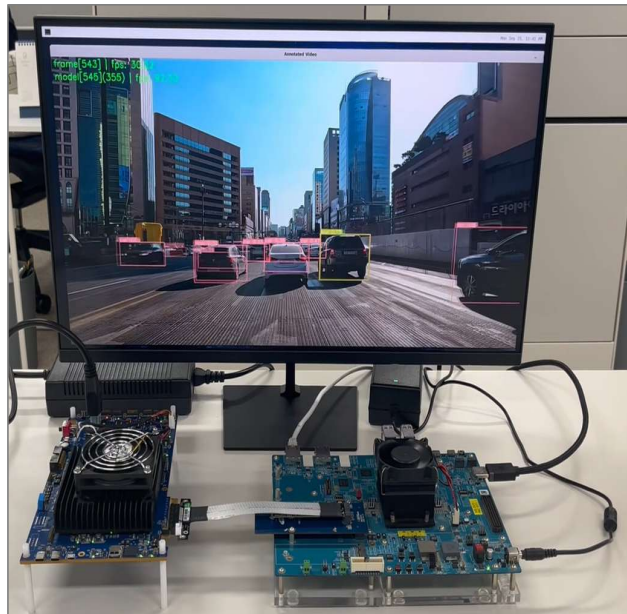
Evaluation board of Eagle-N A0 sample

- LPDDR5x 7.5Gbps 16bit x 8 channel
- 2x PCIe 4-lane Gen5 PCIe with 32GT/s
- QSPI and Octal DDR200
- SD/eMMC with SD 4.2 and eMMC 5.1
- 7x I2C, 6x SPI, 4x UART, 8x PWM, 64x GPIO, JTAG, Alert IO



A0 sample for PoC – AI demo (1/2)

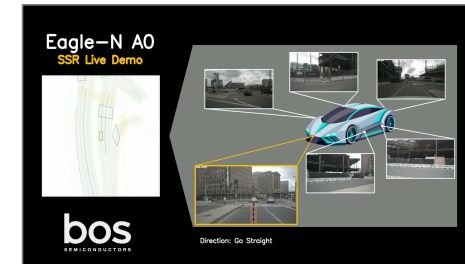
Yolo



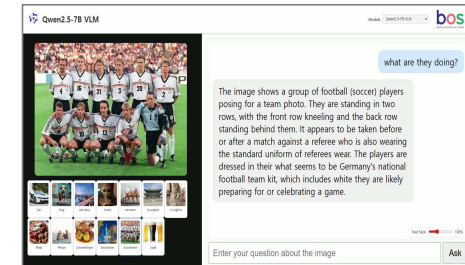
Resnet/VIT



SSR



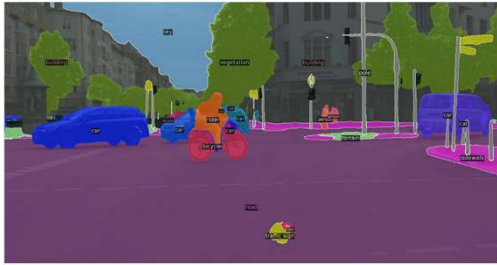
Qwen 2.5-7B, VLM



- **YOLO**: An object detection model that identifies the locations and classes of objects in images or videos in real time.
- **Resnet**: An image classification model that uses residual connections to train deep neural networks effectively.
- **VIT** (Vision Transformer): An image classification model that divides an image into patches and processes them using a Transformer architecture.
- **SSR** (Sparse Scene Representation): An end-to-end autonomous driving model that learns sparse scene representations to integrate perception and driving planning.
- **Qwen-VL**: A multimodal model that jointly processes images and text for visual question answering and document understanding.

A0 sample for PoC – AI demo (2/2)

PDL



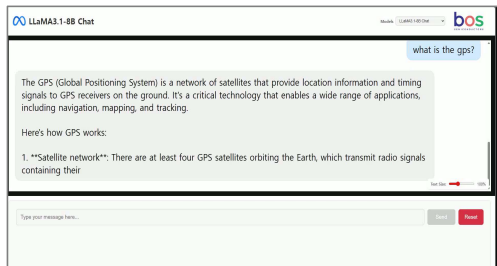
OFT



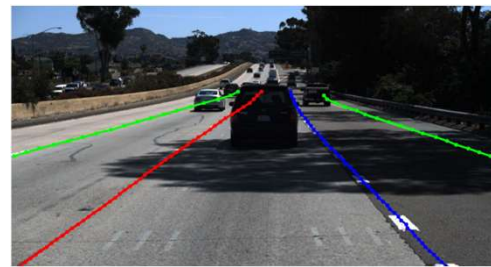
Stable Diffusion



Llama 3.1-8B, LLM



UFLD



HRNet



- **PDL** (Panoptic-DeepLab): A panoptic segmentation model that classifies every pixel while distinguishing individual object instances.
- **Llama**: Meta's large language model for text generation, question answering, summarization, and translation.
- **OFT** (Orthographic Feature Transform): A 3D object detection model that transforms image features into a bird's-eye-view representation.
- **UFLD** (Ultra Fast Lane Detection): A lane detection model that identifies lane regions and separates individual lane instances in road images.
- **Stable Diffusion**: An image generation model that creates or edits images from text descriptions (Prompt: "Tropical beach with palm trees, crystal clear water").
- **HRNet** (High Resolution Network): A vision model for human pose estimation by detecting body keypoints.

AI Model Evaluation (1/2)

- Eagle-N validation against real-world AI workloads demonstrates the following:

Workload	Metric / Configuration	Performance
Generative AI (Llama 8B)	Batch Size 1	17.3 TPS
	4-bit Quantization + Speculative Decoding	50 TPS
Computer Vision (ResNet-50)	Batch Size 4	>6,000 FPS

AI Model Evaluation (2/2)

- Detailed BM results of LLM/VLM models for various model sizes
 - Input tokens: The number of text tokens provided to the model
 - Output tokens: The number of text tokens generated by the model in the response
 - Context size: The maximum sequence length supported by the KV cache, including input and output tokens
 - ViT delay: The latency required for the Vision Transformer to convert an image into visual tokens
- With Eagle-N speculative decoding and 4bit quantization, the evaluated 8B models show ~50 TPS
 - Decode throughput increases as the model size decreases

Model	Parameter s	HW Architecture		Input(Video)	Data type	Input Tokens	Output Tokens	Context Size	ViT Delay (ms)	TTFT (ms)	Output TPS (Tokens/s)
		TOPS	Memory BW								
Qwen2.5	3B	250	204	-	4bit	128	256	4096	-	39.81	107.10
	7B	250	204	-	4bit	128	256	4096	-	72.85	54.80
Qwen2.5-VL	3B	250	204	224x224	4bit	128	256	4096	38.97	80.93	106.20
	7B	250	204	224x224	4bit	128	256	4096	39.75	128.32	54.60
Qwen3	4B	250	204	-	4bit	128	256	4096	-	51.07	91.70
	8B	250	204	-	4bit	128	256	4096	-	81.76	50.50
Qwen3-VL	4B	250	204	256x256	4bit	128	256	4096	15.16	72.32	90.00
	8B	250	204	256x256	4bit	128	256	4096	28.33	122.77	50.20



Thank you.

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